

ECE 143: Programming for Data Analysis

Modeling Predictions on Football Team Builds

Team Code Force

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Methodology

- Database Processing & Setup
- Visualization Questions
 - Experimentation / Novel Visuals
 - Parameters & Game Results
- Prediction Model to build a football team





Data Preprocessing & Setup

- Uses European Soccer Database from Kaggle
- Dataset information covers +25,000 matches, +10,000 players, players and teams' attributes, team line-up, details about each match (goal types, possession, corner, cross, fouls, cards, etc.), and betting odds from seasons 2008 to 2016
- Input: Loaded in database.sqlite with player attributes and team attributes
- Used only player/team records dated on or before each match to prevent data leakage
- Combined all engineered features with match outcomes and betting odds to create a structured ML dataset
- Output: Save file as .npz or .parquet

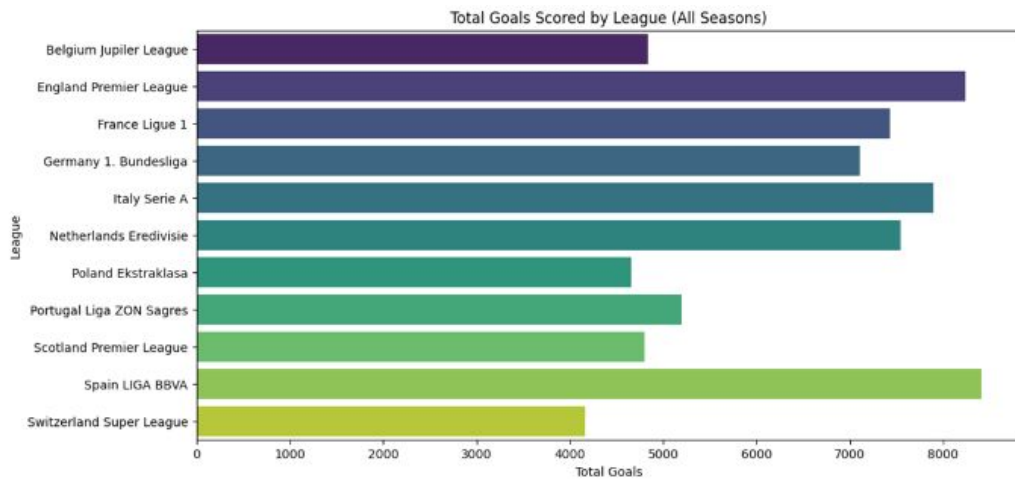


Visualization Questions: Experimentation

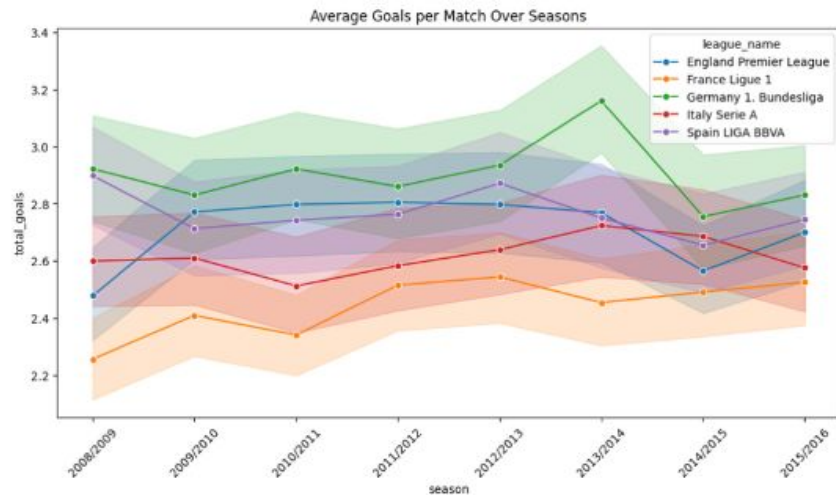
- By analyzing the data, the visualization study aims to uncover the underlying patterns governing match outcomes, player development, and team tactical efficiency
- Performs descriptive analyses on matches, player & team attributes, in-match dynamics (possession & discipline), and advanced dimensionality reduction



Descriptive Match Analysis



Highlights offensive dominance of the Spanish LIGA BBVA and English Premier League and is a baseline for league comparison

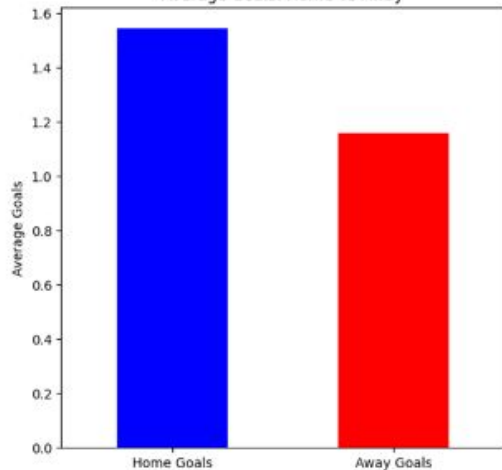


Tracks evolution of scoring and if teams are becoming more offensive or defensive over time



Descriptive Match Analysis

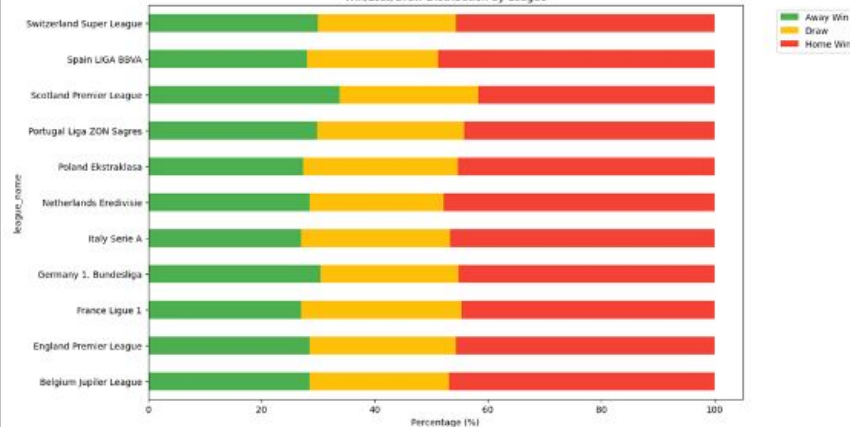
Average Goals: Home vs Away



European Soccer: Home vs. Away Win Rates (All Seasons)

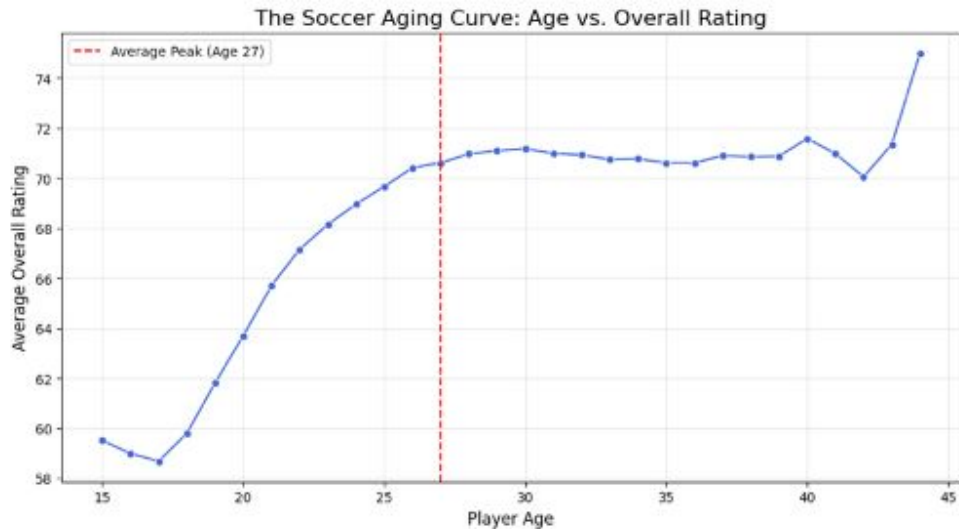


Win/Loss/Draw Distribution by League

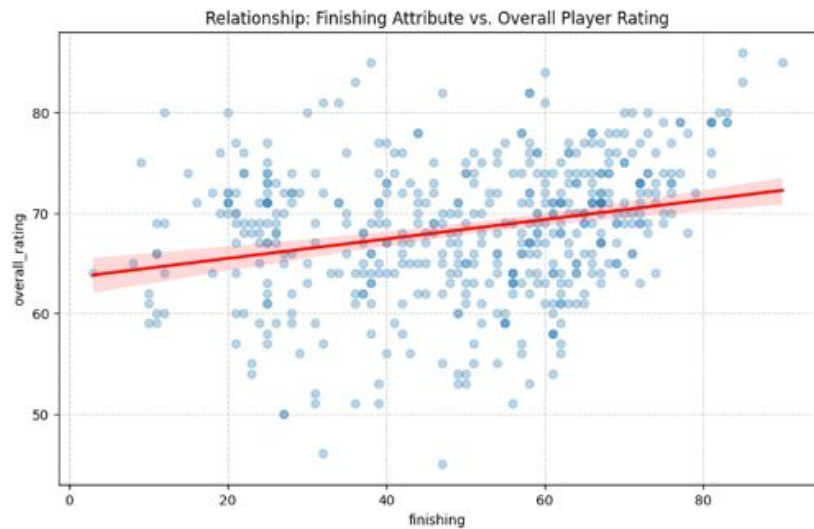


Home Field Advantage: The data confirms that home teams consistently score more and win significantly more often (roughly 45-50% win rate) than away teams across all European territories

Player & Team Attributes

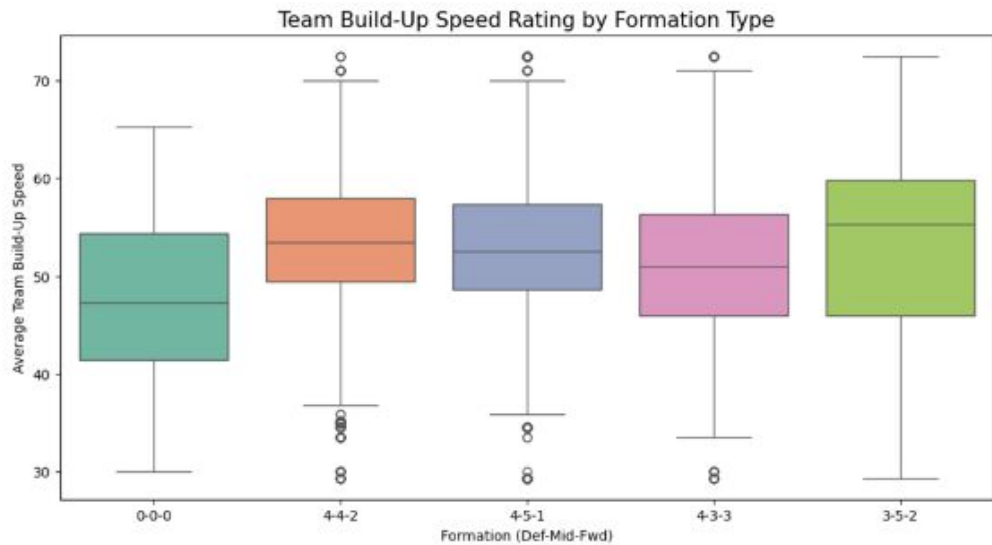
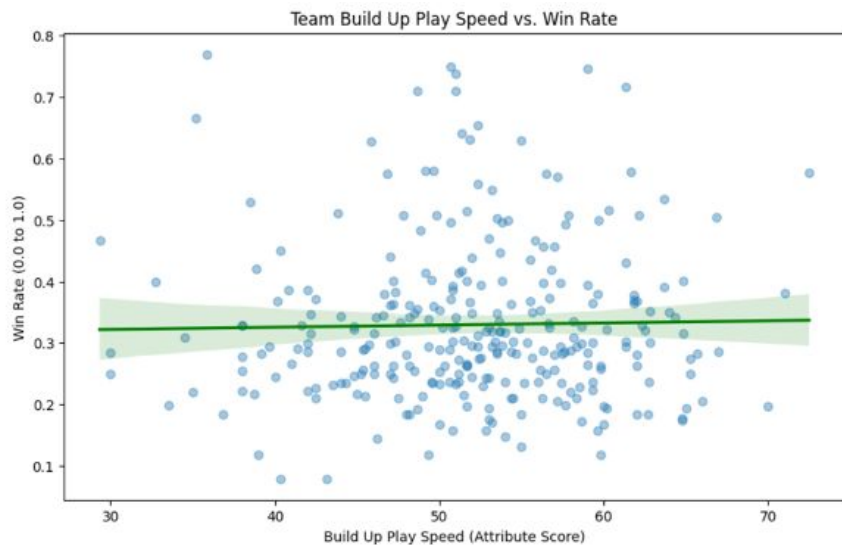


Visualizes physical and technical trajectory of a professional footballer, showing a performance peak between ages 26-29



Confirms high correlation between a player's finishing ability and their total value

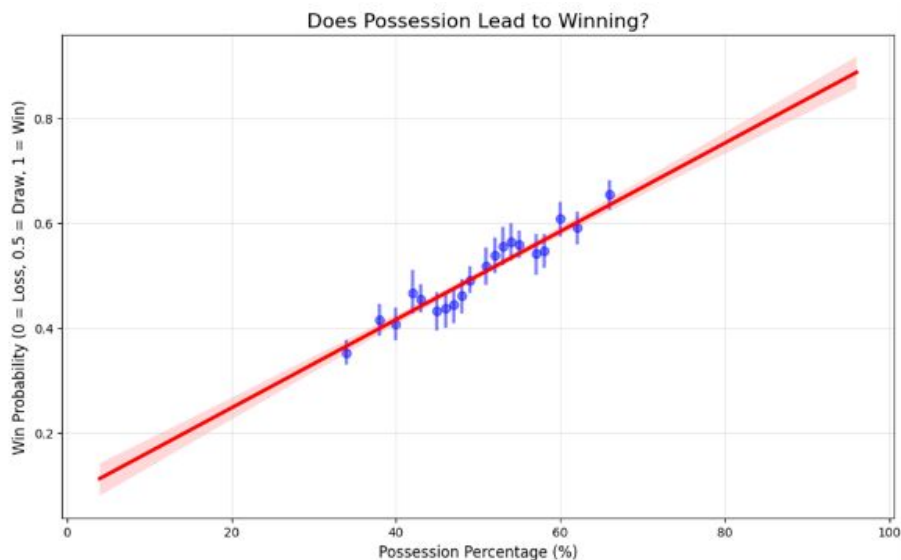
Player & Team Attributes



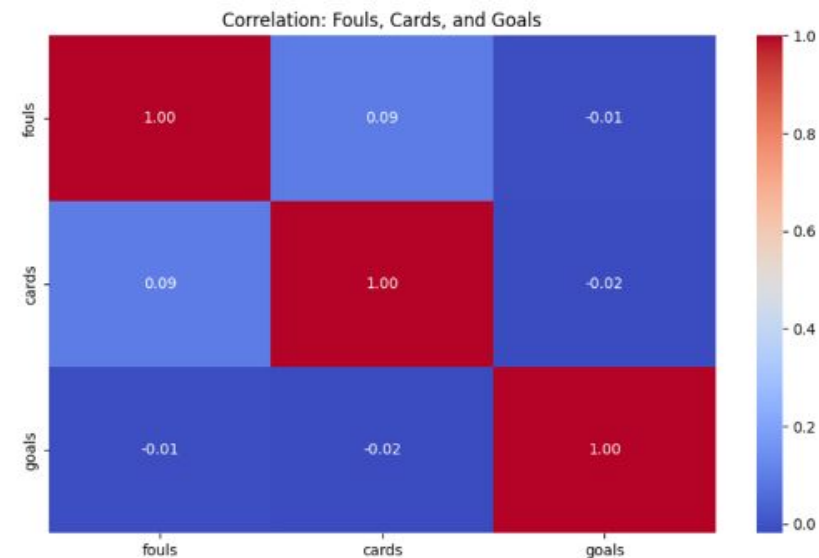
Shows how different tactical setups (like a 4-3-3) correlate with specific player styles (fast vs slow build up) and how these choices impact the team's win percentage



In-Match Dynamics



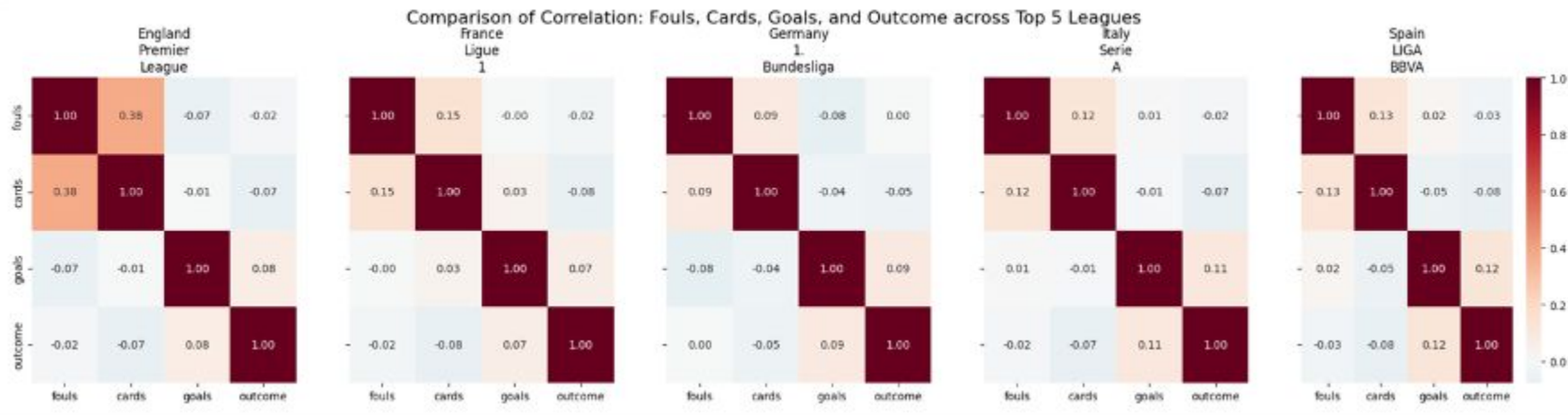
As a team's possession percentage increases, the probability of winning the matches rises linearly, showing importance of ball control



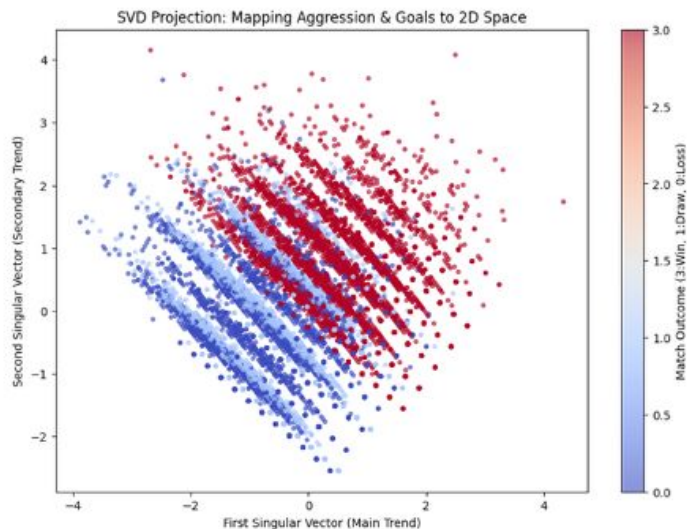
Shows strong relationship between fouls committed and cards received



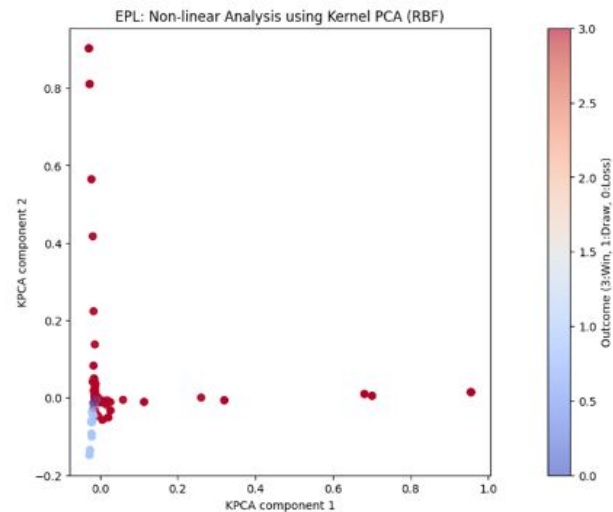
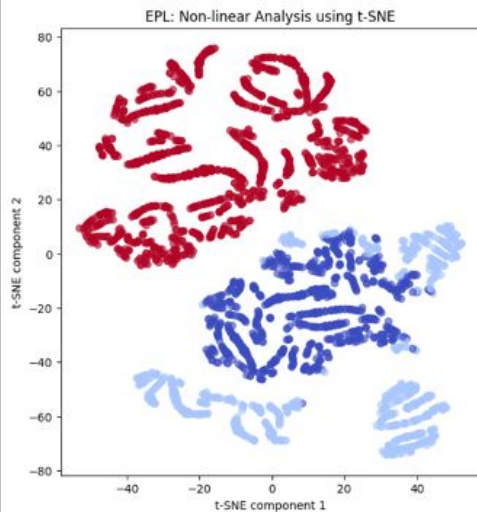
In-Match Dynamics



Advanced Dimensionality Reduction



Collapses match data (goals, fouls, cards) into a 2D plane, attempts to see if wins, losses, and draws separate into distinct spatial zones



Using t-SNE, we mapped complex match features into clusters and revealed local structures, showing that certain types of matches (e.g., "high-intensity draws") cluster together, providing a "fingerprint" of different match types.



Visualization Questions: Experimentation

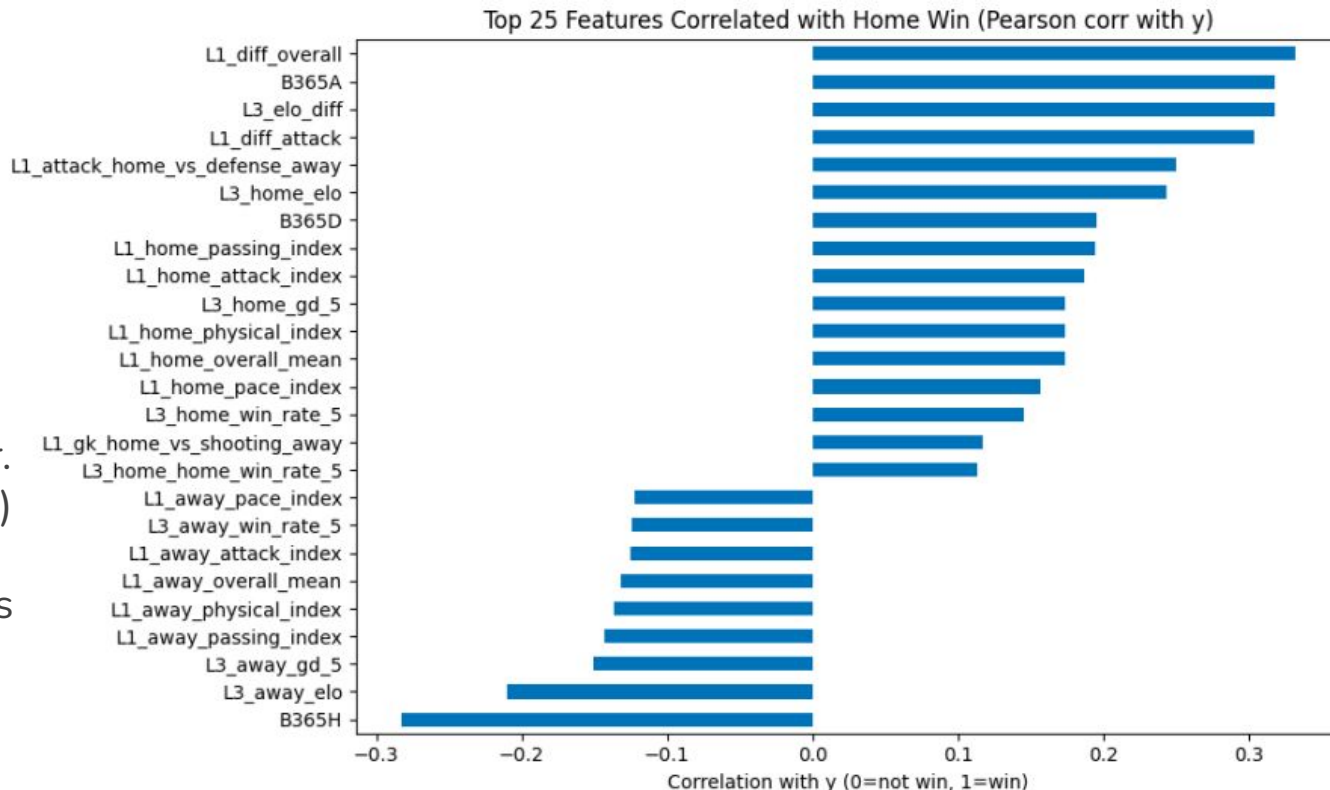
- **Linear vs. Manifold Logic:** While the player & team attributes analysis proves that single factors like possession or fouls influence games linearly, the t-SNE analysis proves that the result of a match is more than the sum of its parts
 - We see that a "win" isn't just caused by high possession; it is a specific combination of possession, low cards, and efficient finishing that creates a "winning cluster"
- **Explaining the Anomalies:** Outlier matches may belong to a "counter-attacking cluster" where high efficiency compensates for low ball control
- **From Statistics to Strategy:** The advanced dimensionality reduction essentially takes the raw in-match dynamics and distills them into a map of "tactical profiles," allowing us to categorize matches not just by score, but by their internal statistical behavior

Visualization Question: Parameter Analysis

Feature-Outcome Correlation

Outcomes

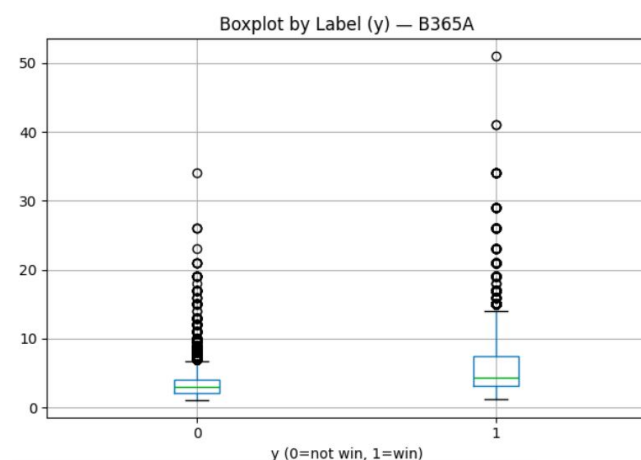
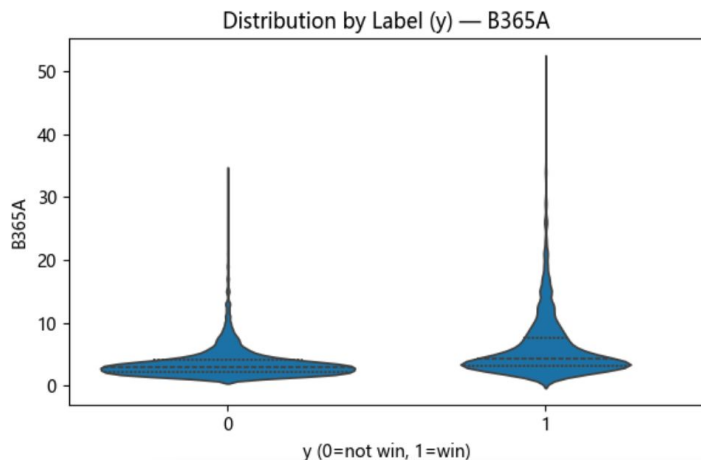
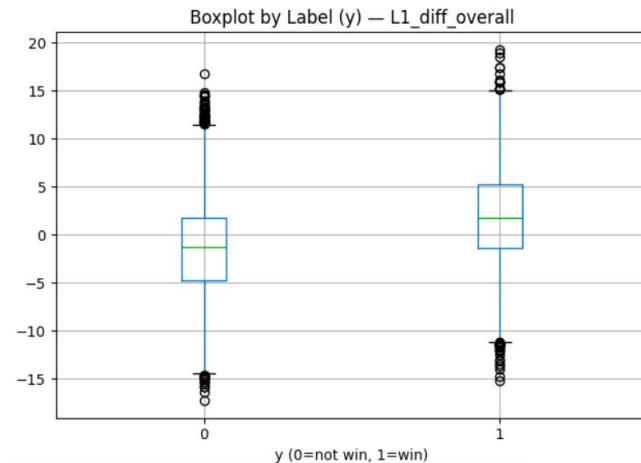
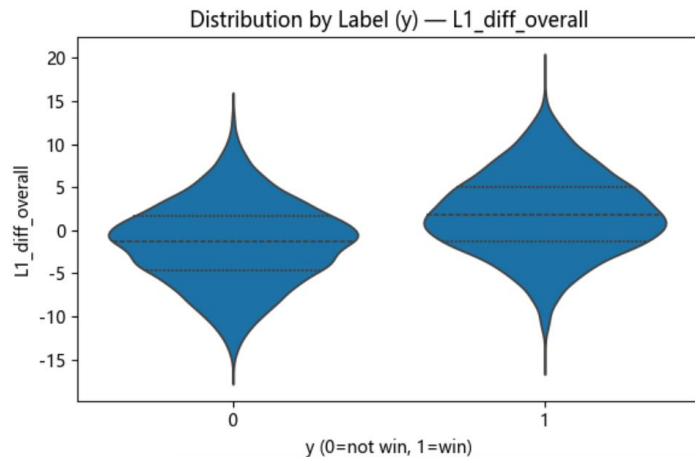
- 1 : Home win
 - 0: Else (including ties)
-
- Positive value: corr. with home wins (1)
 - Negative value: corr. with non-wins (0)





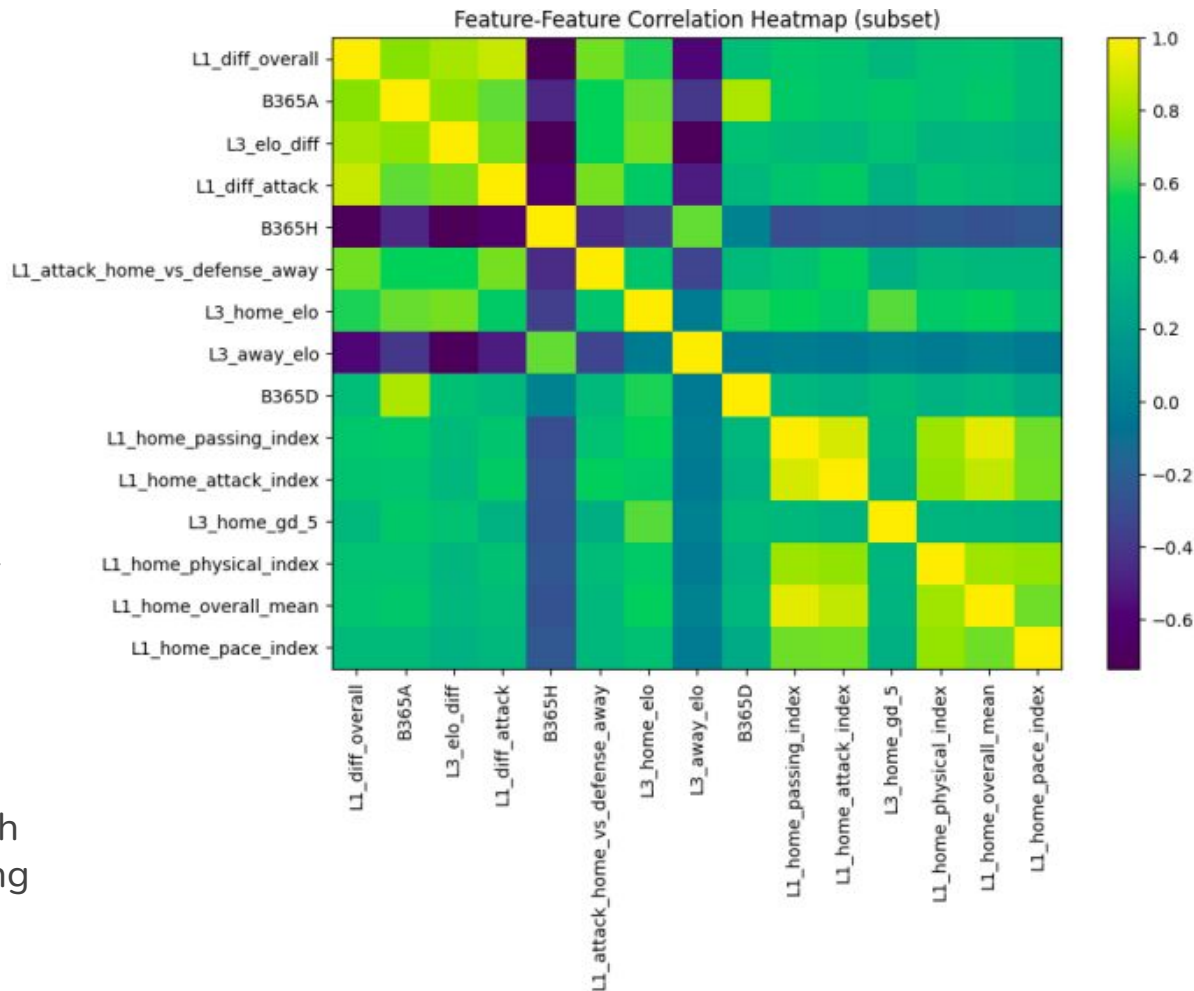
Distribution Analysis within Features

- Grouped by game results: home win/non-win (1/0)
- Comparing one feature **between outcome groups**
 - Median value
 - Overlap
 - Distribution shape
 - Outliers



Pairwise Feature-Feature Correlation

- Strong **positive correlation** indicates similar information/overlap
- Vice versa for **negative correlation** (such as B365H betting odds)





Prediction Modeling

Goal: Predict whether the home team will win a football match.

We aim to analyze how different types of information affect prediction performance, including:

- Player ability (L1)
 - overall, attack, defense, passing, pace, physical, gk, etc
- Team tactical style (L2)
 - buildup play, chance creation, defence, etc
- Recent team performance / Dynamic State (L3)
 - past 5 win rate, goal diff, elo, rest days, etc
- Betting market information

We use **Logistic Regression** as the prediction model.





Prediction Modeling

An **ablation study** is used to evaluate the contribution of each feature group.

Four models are constructed with progressively increasing feature sets:

Model A: Player ability features (L1)

Model B: L1 + Tactical style features (L2)

Model C: L1 + L2 + Recent performance features (L3)

Model D: L1 + L2 + L3 + Betting odds

This layered design allows us to clearly analyze the impact of different information sources.





Prediction Modeling

The dataset is split chronologically into:

- Training, Validation, Test sets

The classification threshold is optimized on the validation set.

Evaluation metrics include:

- AUC, Log-loss, Brier score, Accuracy

Feature engineering:

- Normalization
- Diff. between home/away team features
- Interaction features





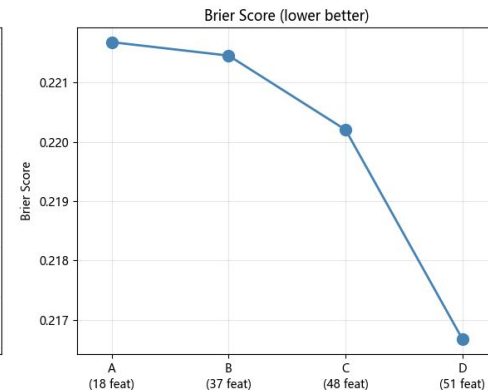
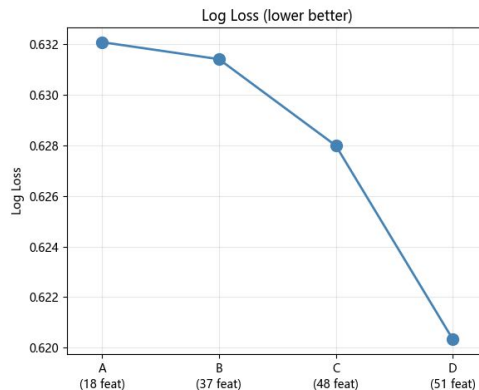
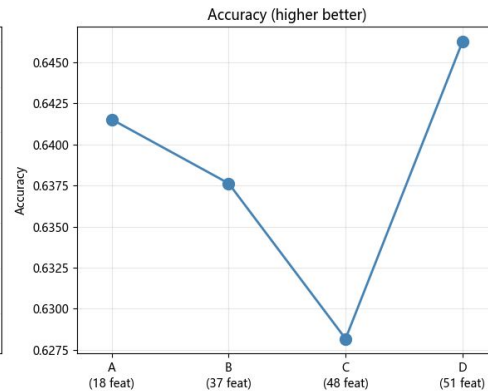
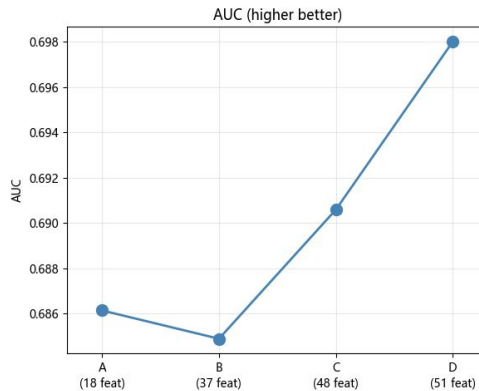
Results & Discussion

Ablation: Model A/B/C/D Metrics

Prediction performance **generally improves** as more information is added.

Impact on predictive performance can be understood through

- Pairwise correlation among feature groups
- Correlation with game results
- Data quality



Thank you!

